



A
Report on

**"Exploratory Data Analysis of
Tesla Stock Market Price History"**

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(SDG Goal No. 8 - Decent Work and Economic Growth)

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At

**DEPARTMENT OF ARTIFICIAL INTELLIGENCE & DATA SCIENCE
AISSMS IOIT Pune**

Affiliated to



SPPU

CERTIFICATE

This is to certify that the report entitled

"Exploratory Data Analysis of Tesla Stock Market Price History"

submitted by students

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is a record of work carried out by them, in partial fulfillment of
requirement for Second Year of Engineering AI-DS at AISSMS IOIT Pune 411001.

This work is done during the academic year 2023-2024, under our guidance.

Date :

Place:

Pune

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ABSTRACT

This project presents a comprehensive **Exploratory Data Analysis (EDA)** of the Tesla Inc. (TSLA) stock price history from 2010 to 2021 using Python. The dataset was sourced from Kaggle and contains daily trading data including Open, High, Low, Close prices, Adjusted Close, and Volume.

Unit I concepts covered include univariate analysis, numerical data types, outlier detection using Z-Score and IQR methods, data transformation, grouping, aggregation, and pivot tables. **Unit II** concepts include histograms, box plots, line charts, candlestick charts, time series visualization, resampling, and grouping of time series data.

Key findings: Tesla stock showed a hockey-stick growth pattern — flat below \$100 for a decade, then surging over 700% in 2020. The stock has high volatility, significant outlier trading days tied to real-world events, and seasonal price patterns visible in the monthly pivot table heatmap.

The project is implemented in Python (Pandas, NumPy, Matplotlib, Seaborn) and runs on Google Colab using the Kaggle CSV dataset.

Keywords

Exploratory Data Analysis, Tesla, Stock Market, Time Series, Univariate Analysis, Outlier Detection, Z-S

Hardware / Software Requirements

Hardware Requirements

Component	Minimum Requirement
Processor	Intel Core i3 or equivalent (1.5 GHz+)
RAM	4 GB (8 GB recommended)
Storage	500 MB free disk space
Internet	Required (for Google Colab and Kaggle download)
Display	1280 x 720 resolution or higher

Software Requirements

Software / Tool	Version / Details
Operating System	Windows 10 / Ubuntu 20.04 / macOS
Platform	Google Colab (cloud-based, free)
Language	Python 3.10+
Pandas	1.5+ — Data loading and manipulation
NumPy	1.23+ — Numerical computations
Matplotlib	3.6+ — Plotting and visualization
Seaborn	0.12+ — Statistical data visualization
Dataset	TSLA.csv from Kaggle (CSV format)

Methodology

The methodology followed is a structured step-by-step EDA pipeline aligned with Unit I and Unit II of the syllabus:

Step 1 — Data Collection

Tesla Stock Price History (TSLA.csv) downloaded from Kaggle. Contains 2,704 rows and 7 columns covering daily stock data from June 2010 to June 2021.

Step 2 — Data Loading and Overview

CSV loaded into a Pandas DataFrame. Shape, data types, date range, and missing values checked. All price/volume columns are numerical (float64 / int64).

Step 3 — Data Transformation

New columns derived: Year, Month from Date; Daily Return ($\text{pct_change} * 100$); Intraday Range (High - Low); 7-Day MA and 30-Day MA (rolling mean); 30-Day Rolling Volatility (rolling std of Daily Return).

Step 4 — Outlier Detection

Z-Score method flags $|Z| > 3$ as outliers. IQR method flags values outside $Q1 - 1.5*IQR$ and $Q3 + 1.5*IQR$. Outliers identified but NOT removed as they represent real financial events.

Step 5 — Univariate Analysis

Close Price, Daily Returns, Volume, and Volatility studied individually using histograms and box plots to understand distribution, shape, and spread.

Step 6 — Data Aggregation and Pivot Table

Data grouped by Year using `groupby()` for yearly summary stats. `pivot_table()` creates Year x Month matrix of average Close prices. Seaborn heatmap visualizes the pivot table.

Step 7 — Time Series Analysis

Full price history line chart; Moving average overlay; Candlestick chart (last 90 days); Monthly resampling; Yearly grouped bar chart; Rolling volatility area chart.

Step 8 — Conclusions

Printed conclusion below each plot interprets findings in the context of Tesla market behaviour and real-world events.

Modules

The project is divided into 6 modules, each corresponding to a section in the Jupyter Notebook:

Module 1 — Data Loading and Overview

Loads TSLA.csv, converts Date to datetime, sorts records, and prints shape, data types, date range, and missing value counts. Generates a statistical summary (describe table) of all numeric columns.

Output: Output: DataFrame info table, describe() summary.

Module 2 — Data Transformation

Adds columns: Year, Month, Month_Name, Daily_Return (pct_change * 100), Intraday_Range (High - Low), MA_7, MA_30, and Volatility (30-day rolling std of Daily_Return).

Output: Output: Updated DataFrame with 8 new columns.

Module 3 — Outlier Detection

Applies Z-Score method (threshold = plus/minus 3) and IQR method (1.5 * IQR fence) on the Volume column. Prints count and list of outlier trading days for each method.

Output: Output: Outlier count and dates from both methods.

Module 4 — Univariate Analysis

Plot 1: Close Price Histogram with mean and median lines. Plot 2: Volume Box Plot with outlier dots. Plot 3: Daily Returns Histogram with KDE curve, skewness, kurtosis annotation. Plot 4: Volatility Box Plot with median highlighted.

Output: Output: 4 matplotlib charts, each with a printed conclusion below.

Module 5 — Aggregation and Pivot Table

Groups data by Year using groupby() to compute Avg_Close, Max_Close, Min_Close, Avg_Volume. Creates pivot_table() (Year x Month, avg Close). Visualizes pivot as a seaborn heatmap.

Output: Output: Yearly summary table, pivot table, heatmap chart.

Module 6 — Time Series Analysis

Plot 1: Full Close Price line chart (2010-2021). Plot 2: Moving Averages (7D and 30D) on Close Price (2019-2021). Plot 3: Candlestick chart (last 90 trading days). Plot 4: Monthly resampled Close Price. Plot 5: Yearly grouped average bar chart. Plot 6: 30-Day Rolling Volatility area chart.

Output: Output: 6 time series charts with printed conclusions.

Working

The project runs as a Jupyter Notebook on Google Colab. Below is the complete working flow:

1. Open Google Colab at colab.research.google.com.
2. Upload the `tesla_eda_syllabus.ipynb` notebook file via File > Upload Notebook.
3. Run Cell 1 — imports all libraries (Pandas, NumPy, Matplotlib, Seaborn).
4. Run Cell 2 — a file upload button appears. Upload `TSLA.csv` downloaded from Kaggle.
5. Run Cell 3 — loads CSV, converts Date to datetime, sorts oldest to newest.
6. Run Cell 4 — displays shape, data types, date range, missing values.
7. Run Cell 5 — displays statistical summary (describe table).
8. Run Cell 6 — adds all derived columns (Year, Month, Daily_Return, MA_7, MA_30, Volatility, Intraday_Range).
9. Run Cell 7 — outlier detection using Z-Score and IQR on Volume, prints results.
10. Run Cells 8-11 — four Univariate Analysis plots with conclusions printed below each.
11. Run Cells 12-13 — yearly aggregation table and monthly pivot table printed.
12. Run Cell 14 — monthly pivot table heatmap displayed.
13. Run Cells 15-20 — six Time Series Analysis plots with conclusions.
14. Final cell displays the EDA Summary Table mapping plots to syllabus topics.

Plot Rendering

All plots use Matplotlib and Seaborn only. Each chart is rendered with `plt.show()` and the conclusion is printed using `print()` immediately after the plot cell. This keeps the notebook clean and easy to follow.

Features

+ Real Kaggle Dataset

Uses actual Tesla stock price data from Kaggle (2010-2021) — not simulated.

+ Syllabus-Aligned

Every module and plot is mapped directly to Unit I or Unit II of the syllabus.

+ Dual Outlier Detection

Implements both Z-Score and IQR methods to detect outliers in Volume data.

+ Data Transformation

Automatically derives Daily Return, Moving Averages, Rolling Volatility, and Intraday Range from raw data.

+ Pivot Table and Heatmap

Monthly average Close price pivot table visualized as a colour-coded heatmap.

+ 11 Visualizations with Conclusions

Each plot includes a printed conclusion explaining findings in simple language.

+ Time Series Resampling

Converts daily data to monthly frequency using Pandas `resample()` for clearer trends.

+ Candlestick Chart

Pure Matplotlib candlestick chart with green/red color coding for last 90 days.

+ Google Colab Ready

Includes file upload cell using `google.colab.files` — no API keys needed.

+ Beginner-Friendly Code

Every line has a comment. No complex one-liners. Designed for 2nd-year students.

Test Cases

TC	Test Description	Input	Expected Output	Status
01	Load CSV file	TSLA.csv	DataFrame: 2704 rows, 7 cols	Pass
02	Date conversion	String date column	datetime64 dtype	Pass
03	Missing values check	Full DataFrame	Zero missing values in all columns	Pass
04	Daily Return calculation	Close column	pct_change values, NaN on row 1	Pass
05	7-Day Moving Average	Close column	Rolling mean, NaN first 6 rows	Pass
06	Z-Score outlier detection	Volume column	Outlier days with abs(Z) > 3 printed	Pass
07	IQR outlier detection	Volume column	Outlier days outside 1.5*IQR printed	Pass
08	Pivot table creation	Year, Month, Close	Year x Month avg Close matrix	Pass
09	Monthly resampling	Daily Close prices	Monthly last-value series	Pass
10	Candlestick chart	Last 90 rows OHLC	Green/red candles with wicks	Pass
11	Yearly grouped bar chart	Grouped by Year	Bar per year with avg price label	Pass

Advantages / Limitations

Advantages

- + Uses a real Kaggle dataset making the findings genuine and credible.
- + Covers all required Unit I and Unit II syllabus concepts in one project.
- + Completely free to run on Google Colab — no paid tools needed.
- + Simple code with comments — easy for beginners to understand and modify.
- + Dual outlier detection (Z-Score + IQR) is more reliable than a single method.
- + 11 charts cover a wide range of visualization types from the syllabus.
- + Printed conclusions below each chart make report writing easy.
- + Rolling volatility and moving averages provide real financial insight.
- + Monthly heatmap gives an intuitive multi-year price pattern at a glance.
- + Candlestick chart is a professional financial chart used by real analysts.

Limitations

- Dataset covers only up to June 2021 — excludes the 2022 crash and 2023 recovery.
- Only one stock (Tesla) is analyzed — no comparison with other stocks.
- Bivariate and multivariate analysis not included (out of scope for this syllabus unit).
- Candlestick chart is static (Matplotlib) — not interactive like Plotly charts.
- Outliers detected but not removed — effect of removal was not studied.
- No predictive modeling — this is purely descriptive exploratory analysis.
- Requires manual CSV upload on each new Colab session.

Future Scope

This project establishes a strong EDA foundation. The following extensions can be built on top of this work:

Stock Price Prediction using LSTM

LSTM neural networks can be trained on historical Close prices to predict future stock values — a natural next step after EDA.

Bivariate and Multivariate Analysis

Correlation analysis between multiple stocks using scatter plots, correlation heatmaps, and pair plots.

Portfolio Optimization

Markowitz Mean-Variance Optimization to find optimal weights for a multi-stock portfolio including Tesla.

Live Data via yfinance

Replace the static Kaggle CSV with live data fetched using the yfinance library for always up-to-date analysis.

Sentiment Analysis Integration

Correlate Elon Musk tweets and news headlines with Tesla price movements to study the social media effect on stock behaviour.

Risk Analysis — VaR and Sharpe Ratio

Add Value at Risk (VaR) and Sharpe Ratio calculations to quantify risk-adjusted returns.

Interactive Dashboard

Build a Plotly Dash or Streamlit dashboard for interactive exploration with date range filters and zoom controls.

Benchmark vs S&P; 500

Compare Tesla performance against the S&P; 500 index to determine relative outperformance or underperformance.

Conclusion

This project successfully performed a complete Exploratory Data Analysis on the Tesla Stock Price History dataset (2010-2021) sourced from Kaggle, using Python on Google Colab.

All required concepts from **Unit I** (EDA Fundamentals) and **Unit II** (Data Visualization and Time Series Analysis) were covered. Data transformation, outlier detection (Z-Score and IQR), grouping, aggregation, and pivot tables from Unit I were implemented. From Unit II, histograms, box plots, line charts, candlestick charts, time series visualization, resampling, and grouping were all demonstrated with real financial data.

Key conclusions from this EDA:

Tesla remained flat below \$100 for its first decade, then surged 700%+ in 2020 — a classic hockey-stick growth pattern.

Daily returns are approximately normal with fat tails, confirming Tesla is a high-risk, high-reward stock.

Outlier volume days detected by Z-Score and IQR correspond to real events: COVID crash, S&P 500 inclusion, and quarterly delivery reports.

Moving average crossovers (Golden Cross in mid-2020) accurately predicted the start of the major bull run.

Monthly pivot heatmap revealed Q4 months tend to be slightly stronger, likely linked to delivery report announcements.

Rolling volatility confirmed volatility clustering, with the COVID crash causing the largest spike in the full dataset.

Monthly resampling provided a cleaner long-term view of price trends without daily noise.

Overall, this project demonstrates how EDA techniques extract meaningful insights from real-world financial data. The concepts applied here form the foundation for advanced data science tasks such as predictive modeling, risk analysis, and portfolio optimization in future work.

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